THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 1	Each point where the graph meets the x-axis is a zero
2	**(a)** 5	Sum of zeroes $= 7$, so other $= 7 - 2 = 5$
3	**(b)** $x^2 - 7x + 12$	$x^2 - (\text{sum})x + \text{product}$
4	**(d)** 9	$\alpha + \beta = -b/a$
5	**(a)** 14	$\alpha\beta = c/a$
6	**(d)** 0	2 is a zero, so the remainder is 0
7	**(d)** 7	Sum $= -b/a$
8	**(b)** 2	Each point where the graph meets the x-axis is a zero
9	**(c)** 6	Sum of zeroes $= 9$, so other $= 9 - 3 = 6$
10	**(a)** $x^2 - 3x - 10$	$x^2 - (\text{sum})x + \text{product}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Zeroes are -2 and 3 (1); a parabola, so degree 2 (1).	1+1
12	Sum $= -\dfrac{b}{a} = 4$ (1); product $= \dfrac{c}{a} = 3$ (1).	1+1
13	$x^2 - (\frac{1}{4})x + (-1)$ (1) $= x^2 - \frac{1}{4}x - 1$ (1).	1+1
14	Zeroes are 1 and 4 (1); a parabola, so degree 2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Zeroes -2, 3 (1); sum $1 = -b/a$ (1); product $-6 = c/a$ (1).	1+1+1
16	Long division (2); quotient $x^2 + x + 3$, remainder -3 (1).	2+1
17	Divide $x^3 - 3x^2 + 3x - 1$ by $x - 1$ (1); remainder 0 (1); it is a factor (1).	1+1+1
	OR: Zeroes $\frac{1}{2}$, 4 (1); sum $\frac{9}{2} = -b/a$ (1); product $2 = c/a$ (1).	1+1+1
18	Long division (2); quotient $2x^2 - 3x + 6$, remainder -10 (1).	2+1
	OR: Divide $x^3 + 2x^2 - 5x - 6$ by $x - 2$ (1); remainder 0 (1); it is a factor (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Quotient $x - 3$, remainder $4x$ (3); dividend = divisor × quotient + remainder verified (2).	3+2
	OR: Sum of zeroes $= -b/a = 8$ (2); third zero $= 8 - (1) - (2) = 5$ (2); check by product $= 10$ (1).	2+2+1
20	Quotient $2x + 3$, remainder $-2x - 5$ (3); dividend = divisor × quotient + remainder verified (2).	3+2
	OR: Sum of zeroes $= -b/a = 6$ (2); third zero $= 6 - (-1) - (3) = 4$ (2); check by product $= -12$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 5	$\alpha+\beta=-b/a$
2	**(c)** -24	$\alpha\beta=c/a$
3	**(b)** 0	4 is a zero, so the remainder is 0
4	**(d)** 7	Sum $=-b/a$
5	**(d)** 3	Each point where the graph meets the x-axis is a zero
6	**(b)** 4	Sum of zeroes $=3$, so other $=3-{-1}=4$
7	**(b)** x^2-9x+8	$x^2-(\text{sum})x+\text{product}$
8	**(a)** 9	$\alpha+\beta=-b/a$
9	**(c)** 20	$\alpha\beta=c/a$
10	**(c)** 0	5 is a zero, so the remainder is 0

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sum $=-\frac{b}{a}=-\frac{5}{3}$ (1); product $=\frac{c}{a}=-\frac{2}{3}$ (1).	1+1
12	$x^2-(2)x+(\frac{1}{3})$ (1) $=x^2-2x+\frac{1}{3}$ (1).	1+1
13	Zeroes are -3 and 1 (1); a parabola, so degree 2 (1).	1+1
14	Sum $=-\frac{b}{a}=1$ (1); product $=\frac{c}{a}=-\frac{3}{4}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Zeroes -5, 2 (1); sum $-3=-b/a$ (1); product $-10=c/a$ (1).	1+1+1
16	Long division (2); quotient x^2-x+3, remainder -2 (1).	2+1
17	Divide x^3+x^2+x+1 by $x+1$ (1); remainder 0 (1); it is a factor (1).	1+1+1
	OR: Zeroes $\frac{3}{2}$, -2 (1); sum $-\frac{1}{2}=-b/a$ (1); product $-3=c/a$ (1).	1+1+1
18	Long division (2); quotient $3x^2-4x+1$, remainder -1 (1).	2+1
	OR: Divide $2x^3-5x^2+x+2$ by $x-2$ (1); remainder 0 (1); it is a factor (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Quotient x^2+1, remainder -4 (3); dividend = divisor × quotient + remainder verified (2).	3+2
	OR: Sum of zeroes $=-b/a=3$ (2); third zero $=3-(2)-(-2)=3$ (2); check by product $=-12$ (1).	2+2+1
20	Quotient x^2+2x-3, remainder $5x+4$ (3); dividend = divisor × quotient + remainder verified (2).	3+2
	OR: Sum of zeroes $=-b/a=4$ (2); third zero $=4-(1)-(-3)=6$ (2); check by product $=-18$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 7	Sum $= -b/a$
2	**(b)** 4	Each point where the graph meets the x-axis is a zero
3	**(a)** 7	Sum of zeroes $= 11$, so other $= 11 - 4 = 7$
4	**(b)** $x^2 - 3x - 18$	$x^2 - (\text{sum})x + \text{product}$
5	**(b)** -4	$\alpha + \beta = -b/a$
6	**(b)** -12	$\alpha\beta = c/a$
7	**(a)** 0	3 is a zero, so the remainder is 0
8	**(c)** 8	Sum $= -b/a$
9	**(d)** 0	Each point where the graph meets the x-axis is a zero
10	**(b)** 5	Sum of zeroes $= 2$, so other $= 2 - -3 = 5$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$x^2 - (\frac{3}{2})x + (1)$ (1) $= x^2 - \frac{3}{2}x + 1$ (1).	1+1
12	Zeroes are 2 and 6 (1); a parabola, so degree 2 (1).	1+1
13	Sum $= -\dfrac{b}{a} = -2$ (1); product $= \dfrac{c}{a} = \frac{1}{5}$ (1).	1+1
14	$x^2 - (-4)x + (4)$ (1) $= x^2 + 4x + 4$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Zeroes $6, -1$ (1); sum $5 = -b/a$ (1); product $-6 = c/a$ (1).	1+1+1
16	Long division (2); quotient $x^2 + 2x + 1$, remainder 4 (1).	2+1
17	Divide $x^3 - 8$ by $x - 2$ (1); remainder 0 (1); it is a factor (1).	1+1+1
	OR: Zeroes $-\frac{1}{3}$, 3 (1); sum $\frac{8}{3} = -b/a$ (1); product $-1 = c/a$ (1).	1+1+1
18	Long division (2); quotient $4x^2 - 7x + 9$, remainder -3 (1).	2+1
	OR: Divide $x^3 + 3x^2 + 3x + 1$ by $x + 2$ (1); remainder -1 (1); it is not a factor (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Quotient $3x^2 - x - \frac{3}{2}$, remainder $\frac{5}{2}x - \frac{1}{2}$ (3); dividend = divisor × quotient + remainder verified (2).	3+2
	OR: Sum of zeroes $= -b/a = 4$ (2); third zero $= 4 - (-1) - (-2) = 7$ (2); check by product $= 14$ (1).	2+2+1
20	Quotient $x^2 - 2x + 1$, remainder 3 (3); dividend = divisor × quotient + remainder verified (2).	3+2
	OR: Sum of zeroes $= -b/a = 3$ (2); third zero $= 3 - (2) - (5) = -4$ (2); check by product $= -40$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.